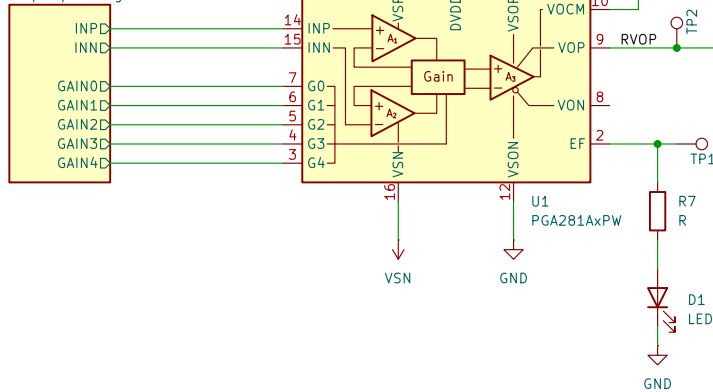


PGA281 gain needs to be 1 lower of the MN2020 ?

Preamp Input Stage



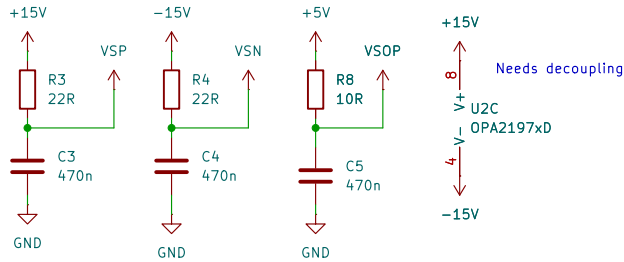
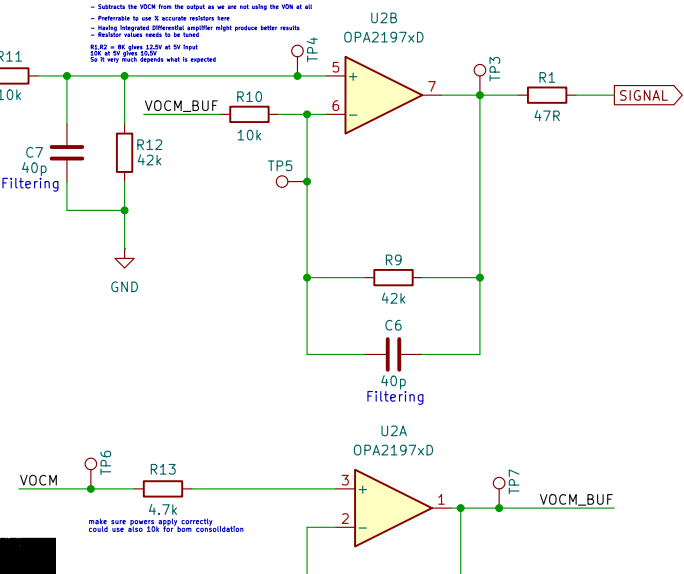
Designer Note:
VSP to DVDD <= MAX 6V

RC-Filter
As defined by the PGA281 datasheet.
Approx 150-200kHz

Differential Amplifier

U2B is at the maximum range of this output stage.
Using the OPA2197, the range is 30V instead of 30V.
The AC input impedance is 100MΩ, etc.
The AC input impedance is 100MΩ, etc.
Subtracts the VOCM from the output as we are not using the VON at all.
Preferable to use 1% accurate resistors here.
Resistor values need to be tuned.

OPA2197xD



PERFORMANCE SPECIFICATIONS ZAD2736 LOW DISTORTION 16-BIT SAMPLE

ANALOG INPUT	
Full-Scale Range (FSR) ^a	±5 V, or 0 to +10 V ^a
Input Impedance	100 MΩ/10 pF
Reference Input Voltage	−6.4 V
Reference Input Impedance	2.7 kΩ

AC CHARACTERISTICS	
Conversion Rate:	
ZAD2736-1, ZAD2736-1B	125 kHz ²
ZAD2736-2, ZAD2736-2B	200 kHz
A/D Conversion Time:	
ZAD2736-1, -1B	5 μs max
ZAD2736-2, -2B	5 μs max
S/H Acquisition Time	3 μs max
S/H Slew Rate	10 V/μs
S/H Aperture Delay	8 ns typical, 12 ns max
S/H Aperture Uncertainty	±0.2 ns, ±0.4 ns, max
S/H Feedthrough	−96 dB typical, −90 dB max
Harmonic Distortion up to 108 kHz	±0.005%

MN2020

DYNAMIC CHARACTERISTICS			
Small Signal Bandwidth			MHz
G=1	5		
G=128	40		
Full Power Bandwidth (FSR=1)	100		
Slew Rate	12		V/μs
Output Settling Time to 10% 25V Step (More 7)			
G=1	5	75	μs
G=128	65		μs

Velho

Sheet: /

File: pga281-ex1.kicad_sch

Title: Preamp PGA281

Size: A4 Date: 2026-07-16

KiCad E.D.A. 10.0.5

Rev: A

Id: 1/2

